Area of Composite Figures

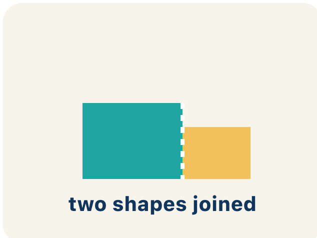
Lesson 5-5

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



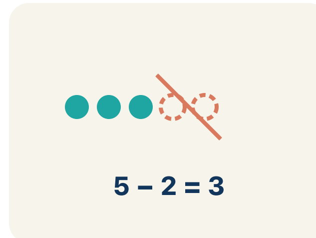
An L-shaped room = a 12×8 rectangle joined to a 6×5 rectangle; total area = 96 + 30 = 126 sq ft

Composite Figure

Write the definition:

____

____



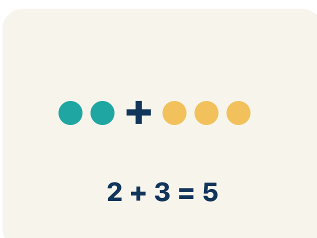
Draw a dashed line across the L-shape to split it into two rectangles — now you can find each area separately

Decompose

Write the definition:

____

____



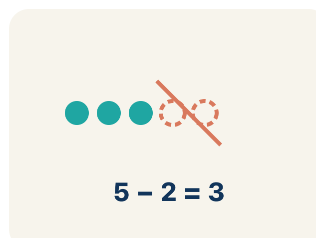
T-shaped hallway: top rectangle = 30 sq ft, bottom rectangle = 28 sq ft → total = 30 + 28 = 58 sq ft

Add

Write the definition:

____

____



A 14×10 pool with a 6×4 cutout: 140 − 24 = 116 sq ft of water surface

Subtract

Write the definition:

____

____

$$3x + 5$$

no equal sign

Rectangle: $A = l \times w$; Triangle: $A = \frac{1}{2} \times b \times h$; use the right formula for each piece of a composite figure

Formula

Write the definition:

____

____

Guided Notes

Level 2 Standard

🎯 **WHAT WE'RE LEARNING TODAY**

I can find the area of a composite figure by adding or subtracting the areas of basic shapes.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Composite Figure | **Decompose** | **Add** | **Subtract** | **Formula**

1. A shape made of two or more basic shapes combined is a ______________ .

2. To break a composite figure into basic shapes is to ______________ it.

3. To find the total area of separate parts, I ______________ their areas.

4. To find the area of a shape with a piece removed, I ______________ the missing part's area.

5. A math rule written with symbols is a ______________ .

🧮 Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: An L-shaped room is made of two rectangles: 10 ft × 6 ft and 4 ft × 3 ft. What is the total area?

A. 72 sq ft

B. 60 sq ft

C. 12 sq ft

D. 72 ft

Step 1 Area 1 = 10 × 6 = 60 sq ft.

Step 2 Area 2 = 4 × 3 = 12 sq ft.

Step 3 Total = 60 + 12 = 72 sq ft.

✅ **Answer:** A. 72 sq ft

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: A rectangular patio is 15 ft × 10 ft with a 5 ft × 4 ft rectangular flower bed cut out. What is the remaining area?

A. 130 sq ft

B. 150 sq ft

C. 20 sq ft

D. 170 sq ft

Step 1 ______________________________

Step 2 ______________________________

Step 3 ______________________________

Answer: ______________________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Find the area of an L-shape made of a 5×3 rectangle and a 2×4 rectangle.

A. 23 sq units

B. 15 sq units

C. 8 sq units

D. 20 sq units

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Plan 2 — L-shaped lobby: decompose the L into two rectangles, one 9 ft by 6 ft and one 7 ft by 3 ft, then ADD. What is the total floor area?

A. 75 sq ft

B. 63 sq ft

C. 54 sq ft

D. 21 sq ft

Show your work:

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2. Plan 4 — Design studio: the floor is a 16 ft by 12 ft rectangle, but a 5 ft by 4 ft storage closet is cut out of one corner. Find the usable floor area.

A. 172 sq ft

B. 192 sq ft

C. 212 sq ft

D. 20 sq ft

Show your work:

Stretch Your Thinking Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A T-shaped room can be decomposed in two different ways. Show both ways and prove they give the same total area. Use a T-shape where the top is 14 ft × 4 ft and the stem is 6 ft × 10 ft.

Sentence starter: Way 1: I split it into ____ and ____. Areas: ____ + ____ = ____. Way 2: I used a ____ rectangle and subtracted ____. Areas: ____ − ____ = ____. Both give ____ sq ft.

Show your work:

Reflect — Exit Ticket

A composite figure is made of a 9 ft x 7 ft rectangle and a 3 ft x 4 ft rectangle joined together. What is the total area?

A. 75 sq ft
B. 63 sq ft
C. 12 sq ft
D. 75 ft

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Composite Figure
2. **Notes 2:** Decompose
3. **Notes 3:** Add
4. **Notes 4:** Subtract
5. **Notes 5:** Formula
6. **We Try (notes):** A. 130 sq ft *— Patio = 15 × 10 = 150 sq ft. Cutout = 5 × 4 = 20 sq ft. Remaining = 150 − 20 = 130 sq ft.*
7. **You Try (notes):** A. 23 sq units *— 5 × 3 = 15. 2 × 4 = 8. Total = 15 + 8 = 23 sq units.*
8. **Try It 1:** A. 75 sq ft *— First rectangle = 9 x 6 = 54 sq ft. Second rectangle = 7 x 3 = 21 sq ft. Joined L-shape, so add: 54 + 21 = 75 sq ft.*
9. **Try It 2:** A. 172 sq ft *— Large rectangle = 16 x 12 = 192 sq ft. Cutout closet = 5 x 4 = 20 sq ft. A piece is removed, so subtract: 192 - 20 = 172 sq ft.*
10. **Exit Ticket:** A. 75 sq ft *— Area 1 = 9 x 7 = 63 sq ft. Area 2 = 3 x 4 = 12 sq ft. Total = 63 + 12 = 75 square feet.*